

Tool Installation Guide

Audience: Ops engineers taking over platform management (GKE-based). **Scope:** installing all tooling needed before configuring gcloud or connecting to Kubernetes. Covered next in **2. gcloud Setup Guide** and **3. Kubernetes Setup Guide**.

Prerequisite: admin/install rights on your machine.

macOS

What gets installed

- **gcloud** — Google Cloud CLI, used to authenticate and manage GCP resources
- **kubectl** — talks to the Kubernetes API on the cluster
- **gke-gcloud-auth-plugin** — required by kubectl to authenticate against GKE; without it kubectl fails even if gcloud works fine
- **kubectx / kubens** — fast switching between cluster contexts and namespaces
- **fzf** — fuzzy finder, gives kubectx/kubens an interactive picker

How to install

Using Homebrew (install from <https://brew.sh> first if not already present):

```
brew install --cask google-cloud-sdk
brew install kubectl
brew install kubectx
brew install fzf
```

If Homebrew isn't permitted by policy, install gcloud via the official installer instead:

```
curl https://sdk.cloud.google.com | bash
exec -l $SHELL
```

Install the GKE auth plugin:

```
gcloud components install gke-gcloud-auth-plugin
```

Enable it — add to ~/.zshrc or ~/.bashrc:

```
export USE_GKE_GCLOUD_AUTH_PLUGIN=True
```

Verify:

```
gcloud --version
kubectl version --client
kubectx --help
```

Windows

What gets installed

- **gcloud** — Google Cloud CLI, used to authenticate and manage GCP

resources

- **kubectl** — talks to the Kubernetes API on the cluster
- **gke-gcloud-auth-plugin** — required by kubectl to authenticate against GKE; without it kubectl fails even if gcloud works fine
- **kubectl-ctx / kubectl-ns** — the Windows-compatible equivalent of kubectx/kubens (the original is a bash script that doesn't run natively on Windows), installed as kubectl plugins via krew
- **fzf** — fuzzy finder; has native Windows support

How to install

gcloud, via the official installer (GUI, user-level install — a signed Google installer, not a disk image, so it isn't subject to ISO download restrictions):

1. Download from <https://cloud.google.com/sdk/docs/install> and run GoogleCloudSDKInstaller.exe.
2. On the final screen, leave the kubectl component checked.

Alternative via **winget**:

```
winget install --id Google.CloudSDK
```

Alternative via **Chocolatey**:

```
choco install gcloudsdk -y
```

fzf:

```
choco install fzf -y
# or
scoop install fzf
```

kubectl-ctx / kubectl-ns via krew:

```
# 1. Install krew – see https://krew.sigs.k8s.io/docs/user-
guide/setup/install/
# 2. Then install the plugins:
kubectl krew install ctx
kubectl krew install ns
```

Usage becomes kubectl ctx and kubectl ns instead of kubectx/kubens. If WSL2 is available, running the actual kubectx/kubens scripts inside WSL2 is an alternative that keeps the exact same command names.

Install the GKE auth plugin:

```
gcloud components install gke-gcloud-auth-plugin
```

Enable it — add to \$PROFILE:

```
[Environment]::SetEnvironmentVariable("USE_GKE_GCLOUD_AUTH_PLUGIN",
"True", "User")
```

Restart the terminal after setting this.

Verify:

```
gcloud --version
kubectl version --client
kubectl ctx --help
```

What's Next

With tooling installed, move to **2. gcloud Setup Guide** to authenticate and configure gcloud for your project(s).